Hwaran Lee

Contact Information
NAVER AI Lab at NAVER Corp. hwaran.lee@gmail.com hwaranlee.github.io

Research Interests

Broad natural language-related topics including Language Models, Language Understanding and Generation, Dialog Systems, Fact Verification, Ethics for AI, Speech Recognition Systems, Neural Networks and Machine Learning

Education

KAIST Daejeon, South Korea
Ph.D., Electrical Engineering Mar. 2013 – Aug. 2018
Dissertation: *Neural Representations for Speech Recognition and Natural Language Generation*
Advisor: Prof. Soo-Young Lee

KAIST Daejeon, South Korea
B.S., Mathematical Science Feb. 2008 – Aug. 2012
Minor: Financial Engineering Program
Magna Cum Laude

Research and Work Experiences

Research Scientist
- NAVER AI LAB, NAVER Corp., Seongnam, South Korea Mar. 2021 – Present

Research Scientist
- T-Brain, AI Center, SK Telecom, Seoul, South Korea Nov. 2018 – Feb. 2021

Graduate Research Assistant
- Electrical Engineering, KAIST, Daejeon, South Korea Mar. 2013 – Aug. 2018

Undergraduate Researcher
- Brain Science Research Center, Daejeon, South Korea Jun. 2012 – Feb. 2013

Publications

International Journal

[J1] **Hwaran Lee**, Seokhwan Jo, HyungJun Kim, Sangkeun Jung, and Tae-Yoon Kim, "SUMBT+LaRL: Effective Multi-domain End-to-end Neural Task-oriented Dialog System", *IEEE Access*, 9 (2021): 116133-116146.

[J2] Geonmin Kim, **Hwaran Lee**, Bo-Kyeong Kim, Sang-Hoon Oh, and Soo-Young Lee, "Unpaired Speech Enhancement by Acoustic and Adversarial Supervision for Speech Recognition", *IEEE Signal Processing Letters*, (2019): 159-163.

[J3] Ho-Gyeong Kim, **Hwaran Lee**, Geonmin Kim, Sang-Hoon Oh, and Soo-Young Lee, "Rescoring of N-best Hypotheses using Top-down Selective Attention for Automatic Speech Recognition", *IEEE Signal Processing Letters*, (2018): 199-203.

[J4] **Hwaran Lee**, Geonmin Kim, Ho-Gyeong Kim, Sang-Hoon Oh, and Soo-Young Lee, "Deep CNNs Along the Time Axis With Intermap Pooling for Robustness to Spectral Variations", *IEEE Signal Processing Letters* 23.10 (2016): 1310-1314.

[J5] **Hwaran Lee**, Nadeem Iqbal, Wonil Chang, and Soo-Young Lee, "A Calibration Method for Eye-Gaze Estimation Systems Based on 3D Geometrical Optics", *IEEE Sensors Journal* 13, no. 9 (2013): 3219-3225.

[J6] Nadeem Iqbal, **Hwaran Lee**, and Soo-Young Lee, "Smart User Interface for Mobile Consumer Devices Using Model-Based Eye-Gaze Estimation", *IEEE Transactions on Consumer Electronics* 59, no. 1 (2013): 161-166.

International Conference

[C1] Kyungjae Lee, Wookje Han, Seung-won Hwang, **Hwaran Lee**, Joonsuk Park, Sang-Woo Lee, "Plug-and-Play Adaptation for Continuously-updated QA", Findings of the Association for Computational Linguistics: **ACL**, 2022 (to appear).

[C2] John Yoon Young Chung, Wooseok Kim, Kang Min Yoo, **Hwaran Lee**, Eytan Adar, Minsuk Chang, "TaleBrush: Sketching Stories with Generative Pretrained Language Models", In Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems, 2022 (to appear)

[C3] Gi-Cheon Kang, Junseok Park, **Hwaran Lee**, Byoung-Tak Zhang, and Jin-Hwa Kim, "Reasoning Visual Dialog with Sparse Graph Learning and Knowledge Transfer", Findings of the Association for Computational Linguistics: **EMNLP**, 2021.

[C4] **Hwaran Lee***, Jinsik Lee*, and Tae-Yoon Kim, "SUMBT: Slot-Utterance Matching for Universal and Scalable Belief Tracker", *The 57th Annual Meeting of the Association for Computational Linguistics* (**ACL**), 2019.

[C5] Geonmin Kim, **Hwaran Lee**, Bo-Kyeong Kim, and Soo-Young Lee, "Compositional Sentence Representation from Character within Large Context Text", *International Conference on Neural Information Processing* (**ICONIP**), 2017.

[C6] Ho-Gyeong Kim, Jihyeon Roh, **Hwaran Lee**, Geonmin Kim, and Soo-Young Lee, "Active Learning for Large-scale Object Classification: from Exploration to Exploitation" *In Proceedings of the 3rd International Conference on Human-Agent Interaction*, (**HAI**), 2015.

[C7] Bo-Kyeong Kim, **Hwaran Lee**, Jihyeon Roh, and Soo-Young Lee, "Hierarchical committee of deep CNNs with exponentially-weighted decision fusion for static facial expression recognition", *In Proceedings of the 2015 ACM on International Conference on Multimodal Interaction* (**ICMI**), 2015.

International Workshop

[W1] Geonmin Kim*, **Hwaran Lee***, CheongAn Lee, Eunmi Hong, Byunggeun Kim, Soo-Young Lee, "A Deep Chatbot for QA and Chitchat.", The Conversational Intelligence Challenge section on NIPS 2017 Competition Track Workshop, 2017.

[W2] **Hwaran Lee**, Geonmin Kim, Jihyeon Roh, and Soo-Young Lee, "Learning Tonotopically Organized Auditory Feature-map from Speech by an Intermap Pooling Layer in a Deep CNN", *15th China-Japan-Korea Joint Workshop on Neurobioloby and Neuroinformatics* (**NBNI**), 2015. (only abstract)

[W3] Geonmin Kim, **Hwaran Lee**, Jaemyung Yu, and Soo-Young Lee, "Spoken Sentence Embedding from Character by Jointly Learning Character-level Compositional Word Model and RNN Sentence Encoder", *15th China-Japan-Korea Joint Workshop on Neurobioloby and Neuroinformatics* (**NBNI**), 2015. (only abstract)

*these authors contributed equally to this work

Patents

[P1] Tae-Yoon Kim, Jin Kim, Hyungjoon Kim, Jinsik Lee, **Hwaran Lee**, Heewon Jeon, Seokhwan Jo,"Method and Apparatus for Providing Hybrid Intelligent Customer Consultation", Korea Patent Application 10-2019-0136035, filed October 2019, Patent Pending.

[P2] **Hwaran Lee**, Jinsik Lee, Tae-Yoon Kim, "Method and Apparatus for Dialogue State Tracking for Use in Goal-oriented Dialog System",

- Korea Patent Application 10-2019-0086380, filed July 2019, Patent Pending.
- PCT/KR2020/008832, filed July 2020, Patent Pending.

Research Projects

Meta Learner Project 2020 - 2021

- Funded by SK Telecom
  - Researched and developed semi-supervised learning algorithms and modules for AutoTrainer
  - Tech leader for research and development of AutoML for various natural language processing tasks

End-to-end Goal-Oriented Dialog Systems 2018 - 2019

- Funded by SK Telecom
  - Researched and developed a scalable multi-domain natural language understanding
  - Researched and developed end-to-end neural goal-oriented dialog systems
  - Participated in the End-to-End Multi-Domain Task-Completion Task (DSTC8 Track1 Task1 at AAAI 2020) and ranked 6th place

A Deep Chatbot for QA and Chitchat 2017

- Project Leader, Funded by *Institute for Information & Communications Technology Promotion (IITP)*
  - Researched and developed an article-based chatbot that carry on both question-answering and chitchat conversations
  - **Ranked 3rd Place** in the Conversational Intelligence Challenge of NIPS 2017 Live Competition

Deep Learning Based Korean Understanding and Applications 2015 – 2018

- Project Leader, Funded by HANCOM Inc.
  - Researched and developed Korean morpheme, syllable-level language models, sentence and document models
  - Researched and developed continual learning algorithms for language models

Speech Feature Extraction Based on Deep Learning for Continuous Speech Recognition Systems 2013 – 2014

- Project Leader, Funded by LG Electronics
  - Developed convolutional neural networks for acoustic models of continuous English speech
  - Programmed C/C++ and CUDA based on KALDI toolkit

Development of Dialog-based Spontaneous Speech Interface Technology on Mobile Platforms 2013 – 2014

- Funded by Electronics and Telecommunications Research Institute *(ETRI)*
  - Developed deep belief networks and auto-encoders for acoustic models of Korean syllable data
  - Programmed models in C/C++ and CUDA languages

Honors and Awards

- Annual Roll Award, KAIST EE — Apr. 2018
- Ranked 3rd, ConvAI challenge, NIPS 2017 Competition Track Workshop — 2017
- Challenge Winner, ICMI EmotiW2015 — 2015
- Best Paper Award, HAI — 2015
- Qualcomm Innovation Award — 2015
- BK21 Plus Financial Support for Graduates Long Term Training — May. 2014
- KAIST Graduate Scholarship — Mar. 2013 - Aug. 2018
- Australian Endeavour Student Exchange Grant (AUD$ 5000), The University of Queensland — Apr. 2011
- National Excellence Scholarship, KOSAF — Feb.2008 - Feb. 2012

Academic Services

- Program Committee / Reviewer
  - ARR 2021, ACL 2021, EMNLP 2021, COLING 2020
  - NeurIPS 2021, ICLR 2021-2022
  - WWW 2022
- Samsung Humantech Paper Awards Committee — 2020
- Qualcomm Innovation Awards Committee — 2019
- Speech Communication — 2019
- IEEE Transactions on Neural Networks and Learning Systems — 2017 - 2018
- Neural Processing Letters — 2015

Invited Talks

- Introduction to deep learning for dialogue systems
  Inha Univ., Seoul, Oct. 2020
  Yeonsei Univ., Seoul, Jun. 2020
  Sookmyung Women's Univ., Seoul, Oct. 2019
- Toward end-to-end neural dialog systems for multi-domain task completion
  KAIST, Daejeon, Dec. 2019

Teaching Experience

Teaching Assistant

- ChatBot PyTorch Implementation, KB-KAIST AI — Sep. 2017
- EE476 Audio-Visual Perception Models — Spring 2016, 2017
- EE538 Neural Networks — Fall 2015, 2016, 2017

Graduate Mentor

- Undergraduate Mentoring Program, KAIST Counselling Center
  Fall 2014, Spring 2016

Skills

- **Languages** Python, Lua, MATLAB, C/C++, CUDA
- **Libraries** PyTorch, TensorFlow, Torch7, KALDI

References

Prof. Soo-Young Lee (Ph.D. advisor)
Emeritus Professor, School of Electrical Engineering, KAIST — sylee@kaist.ac.kr
Director, KAIST Institute for Artificial Intelligence — http://cnsl.kaist.ac.kr